

# Deepfake Detection & Digital Forensics Platform

## Project Proposal & Design Blueprints

*Hackathon Submission Blueprint*

### 1. Problem Understanding

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The rapid advancement of generative AI has made it increasingly easy to create realistic manipulated images, videos, and audio. Deepfakes can be used to spread misinformation, impersonate individuals, conduct fraud, manipulate public opinion, damage reputations, and create misleading narratives on social media.

Most existing deepfake detection solutions primarily attempt to answer: 'Is this media AI-generated or manipulated?' However, a prediction or probability score alone may not provide sufficient information for real-world verification.

A user may also need to know:

- Where did the media originally come from?
- Has the same media appeared online before?
- Has the media been edited or re-contextualized?
- Which part of the media appears manipulated?
- Does the audio match the video?
- Does the file contain suspicious metadata?
- Are weak or outdated cryptographic algorithms present?
- Is the content being presented in the correct context?

Therefore, our solution goes beyond conventional deepfake classification. We propose a multi-layer digital media investigation platform combining:

AI Detection + File Forensics + OSINT + Source Verification + Evidence Correlation

The objective is not merely to label content as fake or real, but to provide an evidence-backed authenticity assessment.

### 2. Proposed Solution

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We propose a web-based platform where users can upload an image, video, or audio file and receive a comprehensive Media Authenticity & Security Report.

Overall Workflow:

1. Upload Media: High-performance client-side file handler with custom drag-and-drop or select file browser logic with real-time UI loading indicators.
2. File & Metadata Analysis: Extraction of EXIF/metadata, checking for missing tags, and computing structural cryptographic hashes.
3. AI Detection Engine (InceptionResNetV2 + SVM): The platform runs face-focused classification using a pre-trained InceptionResNetV2 deep neural network combined with a Support Vector Machine (SVM) to mathematically compute the authenticity percentage.
4. Manipulation Localization (Face Detection via MTCNN): Employs MTCNN (Multi-task Cascaded Convolutional

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Networks) to isolate, crop, and align human faces from video frames to pinpoint the target region of interest for the AI.

5. Audio/Video Consistency Analysis: Sync checking and voice deepfake analysis.

6. OSINT Investigation: Reverse search of sampled frames to identify earlier copies, context drift, or re-uploads.

7. Evidence Correlation & Unified Risk Score: Integrating metadata, face classification, and OSINT history into a single weighted danger score.

8. Forensic Report Generation: PDF/HTML download outlining all technical findings.

## 3. System Architecture

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### 3.1 User Interface

A simple, modern glassmorphic web application where users can:

- Upload videos and inspect batch results with elapsed time
- View analysis progress (e.g., frame extraction percentage)
- View AI prediction and confidence score
- View batch summary statistics
- View OSINT evidence, security metadata findings, timeline, and generate report (Upcoming)

### 3.2 Backend API

An asynchronous FastAPI backend coordinates the complete analysis pipeline. It manages:

- File uploads and async requests
- MTCNN and TensorFlow classification models
- Target integrations for OSINT search, metadata extraction, and report generation (Upcoming)

### 3.3 AI Detection Engine

The uploaded media is analysed using one or more deepfake and manipulation detection techniques. Instead of depending entirely on a single model, the system uses an AI Detection Ensemble. Different models or detection techniques can analyse different characteristics of the media. Their results are combined to produce a more robust confidence estimate.

- Temporal Sampling: Automatically extracts 5 key frames across the video duration.
- Face-based Classification: Utilizes InceptionResNetV2 + SVM to compute facial synthetic authenticity metrics.

### 3.4 File Forensics Engine

Every uploaded file undergoes static security and forensic analysis. The system examines: file type, MIME type, file structure, metadata, EXIF information, cryptographic hashes, hash algorithms, embedded objects, suspicious characteristics, and metadata inconsistencies. The file is analysed statically and is never executed as code.

### 3.5 OSINT Investigation Engine

When suspicious media is detected, the platform investigates publicly available information. It attempts to identify: similar images, similar videos, earlier versions, original sources, reuploads, related articles, public references, and different versions of the same content. For videos, important frames can be extracted and used for visual similarity searches.

### 3.6 Evidence Correlation Engine

The most important component of the architecture combines multiple independent signals: AI Detection + File Forensics

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+ OSINT Evidence + Context Verification + Audio/Video Analysis to build a Unified Risk Score.